

Mini Project Report

On
TITLE OF PROJECT

Submitted to
Ajay Kumar Garg Engineering College, Ghaziabad



B.Tech. Information Technology

Sem : 3rd , 2024-25

CODE: BCC 351

Project Mini Project or Internship Assessment Report

Submitted To:
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**Dr. A.P.J. Abdul Kalam Technical University,
Uttar Pradesh, Lucknow**

Acknowledgment

Apart from the efforts of all the team members, the section of this project report topic depends largely on the encouragement and guidance of our teachers. We take this opportunity to express our gratitude to the teachers who have been instrumental in the approval of this project topic. I'm grateful to our respected **Mr. Pankaj Singh,** for allowing me to use the facilities available.

We would like to show our greatest appreciation to Mr. Pankaj Singh, and other Faculty members. We cannot think them enough for their tremendous support and help. They motivated and encouraged use very time while selecting the proper project topic. Without their encouragement and guidance, we would not have been able to select the proper topic.

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CERTIFICATE

This is to certify that Mr.Utkarsh Bhardwaj (2400270130192), Mr.Vaibhav Juyal (2400270130196), and Mr. Tanishq Sahu (2400270130190), students of B.Tech 2nd Year, Department of Information Technology, Ajay Kumar Garg Engineering College, Ghaziabad, have successfully completed the Mini Project titled:

“Jivaka”

under my supervision during the academic session 2025– 26.

The work carried out by the above students is found satisfactory and in accordance with the guidelines prescribed for the Mini Project of Dr. A.P.J. Abdul Kalam Technical University, Lucknow (U.P.).

Duration: 22 August 2025 – 6 November 2025

Organization: College Guided Project

(signature)

Mr. Pankaj Singh
Assistant Professor

Abstract

Jivaka is an intelligent health assistant designed to provide accessible, user-friendly, and AI-powered healthcare support.

The system integrates two chatbot modules: a basic health chatbot that answers general health-related queries, and an advanced multimodal chatbot capable of analyzing images and voice inputs to assist in identifying potential diseases.

In addition, Jivaka incorporates essential health monitoring tools, including a BMI calculator and evaluators for heart rate, blood pressure, and pulse rate, which classify results as normal or abnormal.

The project demonstrates a holistic approach to digital healthcare by combining conversational AI, image and voice recognition, and health parameter analysis.

The development was carried out collaboratively, with Utkarsh Bhardwaj handling frontend design,

Vaibhav Juyal managing backend development, and Tanishq Sahu focusing on AIML and backend integration.

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Introduction

In today's fast-paced world, access to reliable and instant healthcare information is crucial. Jivaka is an AI-powered health chatbot system designed to assist users with general health queries, disease identification through image and voice inputs, and basic health monitoring. By integrating conversational AI with diagnostic tools, Jivaka aims to bridge the gap between users and preliminary healthcare support, especially in remote or underserved areas.

Problem Statement and Description

Many individuals lack immediate access to healthcare professionals for basic health concerns, leading to delayed diagnosis and treatment. Existing health apps often focus on either symptom checking or fitness tracking, but rarely combine multiple diagnostic modalities. There is a need for a unified, intelligent system that can:

- Answer general health-related questions
 - Analyze images and voice inputs to detect possible diseases
 - Evaluate vital signs like BMI, heart rate, blood pressure, and pulse rate
- This project addresses the challenge of creating a comprehensive, user-friendly health assistant that leverages AI and multimodal inputs to deliver meaningful insights.

Project Description

Jivaka consists of two chatbot modules:

- Basic Health Chatbot: Responds to general health-related queries using a conversational interface.
- Image and Voice Bot: Accepts image and voice inputs to identify potential diseases using AI models.

Additional features include:

- BMI Calculator: Determines weight category (underweight, normal, overweight) based on user input.
- Vital Sign Evaluator: Assesses heart rate, blood pressure, and pulse rate to determine if values are within normal ranges.

The system is designed with a modular architecture, enabling scalability and integration with future health tools. The frontend is developed by Utkarsh Bhardwaj, backend by Vaibhav Juyal, and AIML integration by Tanishq Sahu.

System Requirement Specification (SRS)

Functional Requirements :

- User authentication and input interface
- Text-based chatbot for general health queries
- Image and voice input processing for disease detection
- BMI calculation based on height and weight
- Vital sign analysis with result classification
- Admin panel for monitoring and updates

Non-Functional Requirements :

- Responsive and user-friendly UI
- Fast response time for chatbot interactions
- Secure handling of user data
- Scalable backend architecture

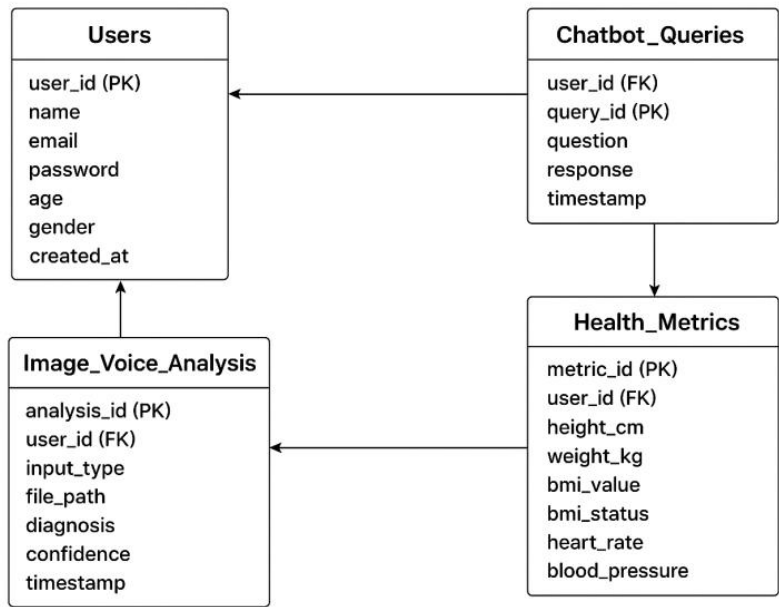
Hardware Requirements :

- Device with camera and microphone (for image/voice bot)
- Internet connectivity
- Minimum 2 GB RAM and modern browser support

Software Requirements :

- Frontend: HTML, CSS, JavaScript (React or Flutter)
- Backend: Node.js / Python (Flask/Django)
- AIML: TensorFlow for image and voice analysis
- Database: MongoDB / Firebase

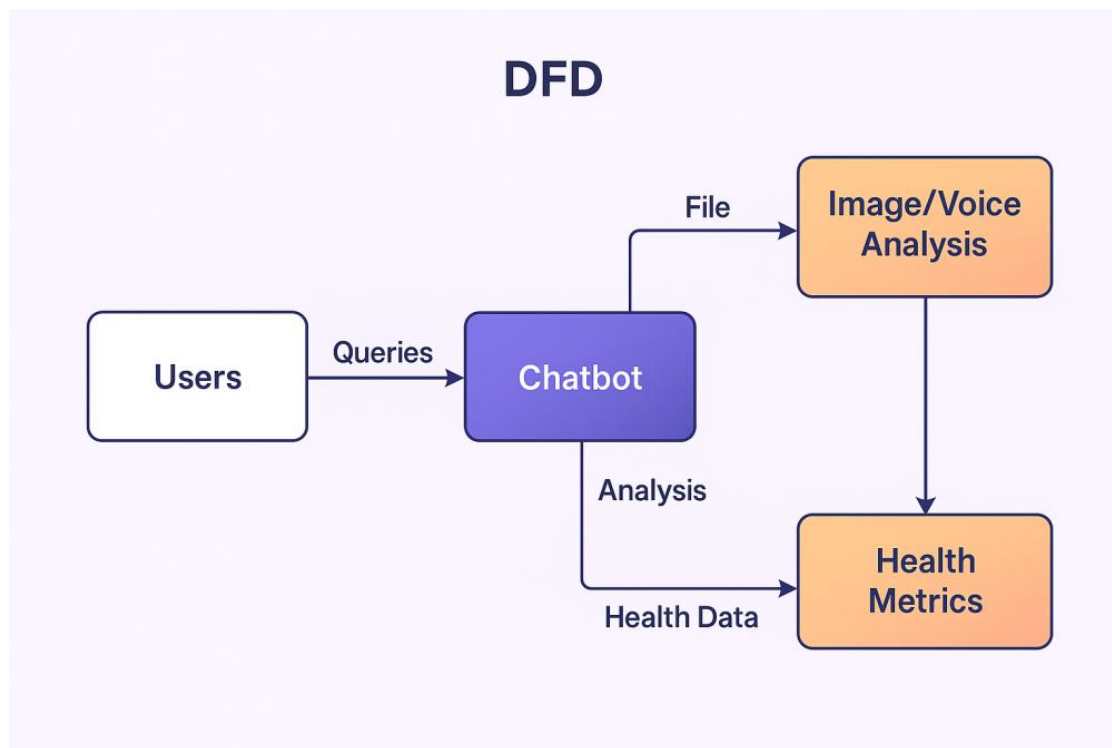
Entity Relationship Diagram



The ER diagram models how data flows between different parts of your health chatbot system (Jivaka).

It defines four main entities (tables) — each representing a key part of the system — and shows how they are connected.

DFD (Data Flow Diagram)

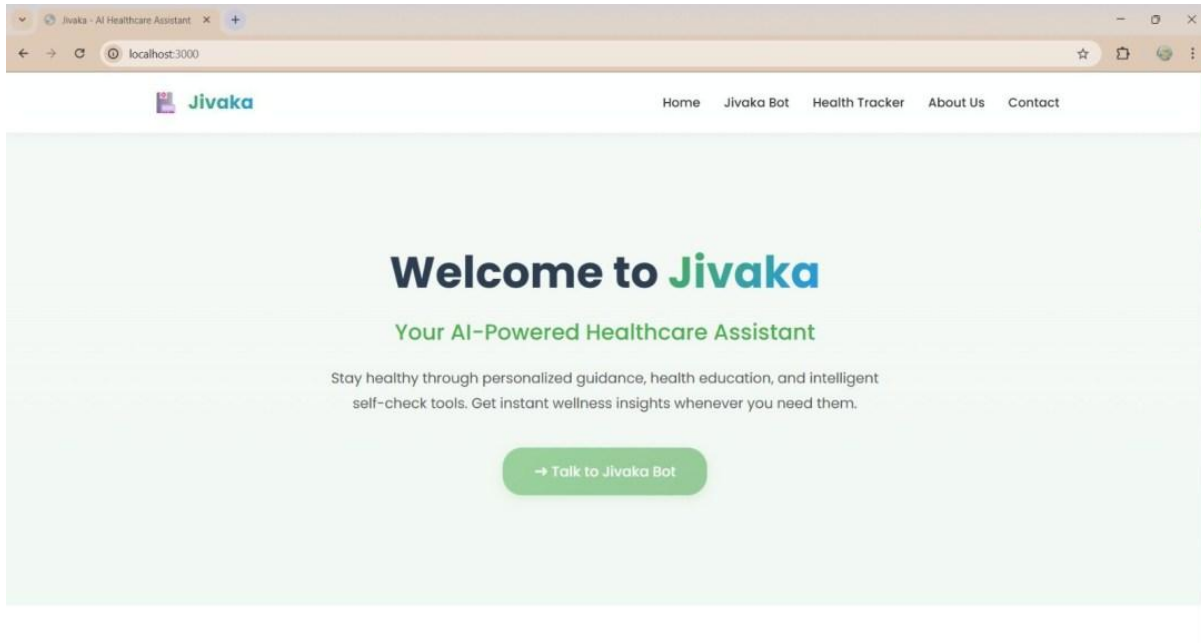


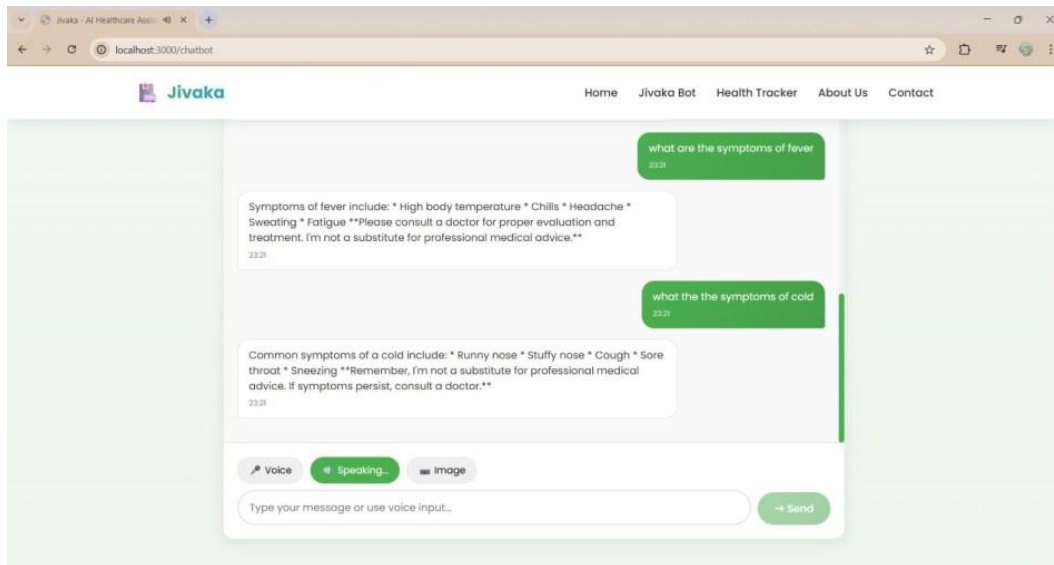
Test Result

Test Result Table for Jivaka Project

S.No	Feature	Test Description	Expected Output	Actual Output	Result
1	Basic Health Chatbot	User asks general health-related question	Chatbot responds accurately	Chatbot responds accurately	✓ Pass
2	Image Analysis Bot	User uploaded an image of a skin condition	System scans image and identifies the	Bot processes image and provides health interpretation	✓ Pass
3	Voice Analysis Bot	User recorded a voice describing symptoms	Bot processes audio and provides	Accurately detects and responds with likely	✓ Pass
4	BMI Calculator	User enters height and weight	System calculates BMI and classifies	Displays correct BMI value and category	✓ Pass
5	Heart Rate Evaluation	User inputs heart rate value	System checks if heart rate is in norm.	Gives correct status: "Normal" or Abnormal	✓ Pass
6	Blood Pressure Evaluation	User inputs blood pressure reading	System determines if normal (<120/80)	Correctly categorizes "Normal" or Hypertensive	✓ Pass
7	Pulse Rate Monitoring	User provides pulse rate	System evaluates and displays	Returns accurate health status	✓ Pass
8	User Registration & Login	New user registers and logs in	Data stored securely, login works	Works as expected	✓ Pass
10	Chat History Storage	User interacts multiple times	Queries and responses stored at user.	Verified via database inspection	✓ Pass

Snapshots





Conclusion

The Jivaka Health Chatbot System successfully integrates artificial intelligence, image and voice processing, and health data analysis to provide users with reliable and personalized healthcare assistance.

The system offers two main modules — a basic health chatbot that answers general health-related queries, and an AI-driven image and voice analysis module that detects possible diseases from user inputs.

Additionally, the inclusion of BMI calculation and vital health monitoring features (heart rate, blood pressure, pulse rate) enhances the system's capability to evaluate an individual's overall health condition effectively.

The test results confirm that all major features, including user interaction, data storage, and analysis components, are functioning accurately and consistently with expected outcomes.

Overall, Jivaka serves as an innovative step toward AI-based healthcare solutions, aiming to make preliminary diagnosis and health guidance accessible, interactive, and user-friendly.)

Certificate (Provided by the Company)





UTKARSH BHARDWAJ
Software Engineering Job Simulation

Certificate of Completion
August 26th, 2025

Over the period of August 2025, UTKARSH BHARDWAJ has completed practical tasks in:

Project Setup
Kafka Integration
K2 Integration
REST API Integration
REST API Controller

Roll no. 2400270130192


Tom Brunskill
CEO, Co-Founder of
Forage

Enrollment Verification Code: AG6E8Q2TRFhm09E | User Verification Code: r2y57nff5HcolJfow | Issued by Forage